Employee Analytics Platform

Data Engineering Architecture Report

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1. Executive Summary

This report provides a comprehensive overview of the data engineering architecture designed to support the Employee Analytics Platform. The system integrates workforce scheduling, certification tracking, and employee reference data into a unified analytical environment, enabling operational visibility and AI-assisted querying through a conversational chatbot interface.

The architecture follows a modern, layered data warehouse pattern built on Snowflake, with automated ingestion pipelines, structured transformations, and a semantic model that powers a Snowflake Cortex Analyst-backed Streamlit application.

2. Architecture Overview

The platform is structured across five distinct layers, each serving a specific role in the data lifecycle. The table below summarises the purpose of each layer.

Layer	Description
Data Sources	Raw input data from Microsoft Outlook, CSV spreadsheets, and employee reference files.
Data Ingestion	GitHub Actions-triggered pipeline executing Python ETL scripts for cleaning and transformation.
Raw Layer	Unprocessed Snowflake tables: Schedule_Raw, Certification_Raw, Employee_Raw.
Curated Layer	Business-ready fact and dimension tables derived from raw data.
Semantic Layer	YAML-based semantic model exposing curated tables to the AI layer.
AI & Application Layer	Snowflake Cortex Analyst + Streamlit chatbot for natural language querying.

3. Data Sources

The platform ingests data from three primary sources:

- **Microsoft Outlook Calendar / Schedule Data —** Employee meeting and calendar events used to derive scheduling metrics and load analysis.
- **Certification Spreadsheet / CSV —** Structured records of employee technology certifications, including validity periods, costs, and expiration statuses.
- **Employee Tech Team Reference CSV —** A canonical employee reference file containing identity, contact, and employment status information.

4. Data Ingestion Pipeline

The ingestion pipeline is automated via GitHub Actions, which serves as the orchestration layer. Upon trigger, the workflow executes a series of Python ETL scripts responsible for the following operations:

- Extracting data from source systems and files.
- Cleaning and standardising raw records (name normalisation, date formatting, deduplication).
- Transforming data into structured formats suitable for loading into Snowflake raw tables.

This approach provides version-controlled, reproducible pipeline execution with full audit trails via Git history.

5. Snowflake Data Warehouse

5.1 Raw Layer

The raw layer stores ingested data in its lightly processed state, preserving source fidelity. Three tables are maintained:

- **Schedule_Raw —** Calendar and meeting event data from Outlook.
- **Certification_Raw —** Technology certification records per employee.
- **Employee_Raw —** Employee identity and employment status records.

5.2 Curated Layer

The curated layer applies business logic, joins, and aggregations to produce analytics-ready tables. The following four tables are defined:

Table Name	Description
FACT_SCHEDULE_MEETINGS	Granular meeting-level records with duration, load percentage, weekday, time of day, load band, and a summary sentence per meeting.
FACT_DAILY_SCHEDULE	Daily aggregated schedule summaries per employee, including total meeting minutes, free time, meeting count, and workload category.
FACT_CERTIFICATIONS	Full certification register per employee with status, expiration, cost, validity years, partner tier, and days until expiry.
DIM_EMPLOYEE	Employee dimension table with identity fields, email, name variants, and employment status for use in joins.

6. Data Model & Relationships

The data model follows a star schema pattern centred on the EMPLOYEE_REFERENCE (DIM_EMPLOYEE) table. The diagram below illustrates the entity relationships, followed by a detailed description of each table.

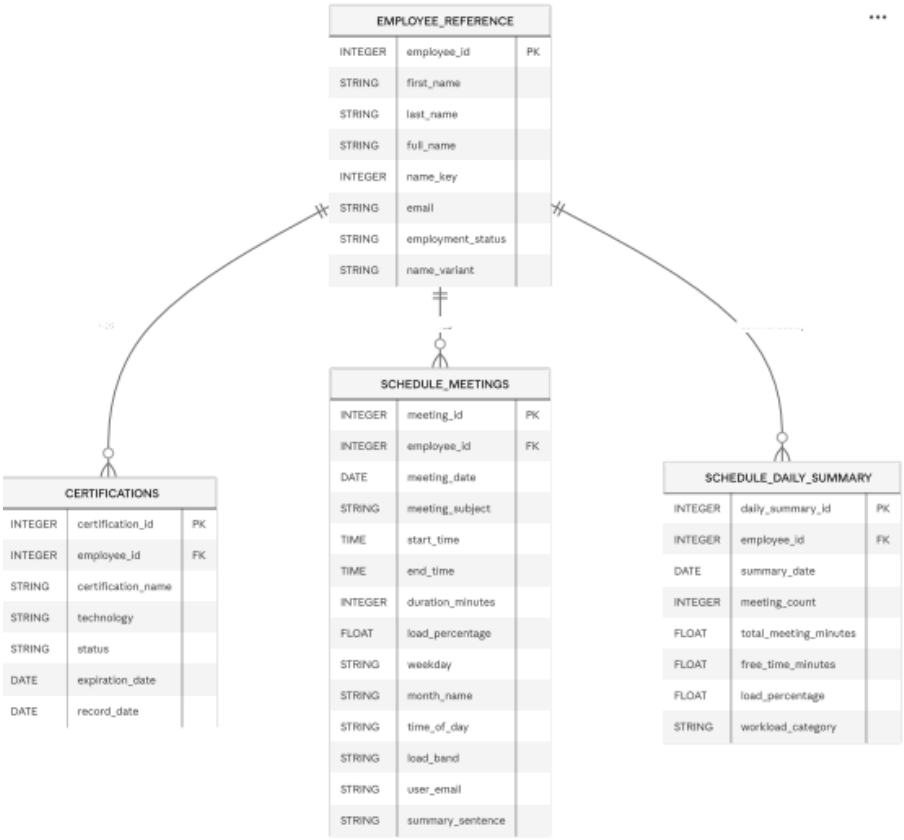

Figure 1: Star Schema — Entity Relationship Diagram

Parent Table	Child Table	Join Key	Cardinality
EMPLOYEE_REFERENCE	SCHEDULE_MEETINGS	employee_id	1 : Many
EMPLOYEE_REFERENCE	SCHEDULE_DAILY_SUMMARY	employee_id	1 : Many
EMPLOYEE_REFERENCE	CERTIFICATIONS	employee_id	1 : Many

6.1 EMPLOYEE_REFERENCE

Acts as the central dimension. **Key fields:** employee_id (PK), first_name, last_name, full_name, name_key, email, employment_status, name_variant.

6.2 SCHEDULE_MEETINGS

Fact table for individual meeting events. **Key fields:** meeting_id (PK), employee_id (FK), meeting_date, meeting_subject, start_time, end_time, duration_minutes, load_percentage, weekday, month_name, time_of_day, load_band, user_email, summary_sentence.

6.3 SCHEDULE_DAILY_SUMMARY

Aggregated daily fact table. **Key fields:** daily_summary_id (PK), employee_id (FK), summary_date, meeting_count, total_meeting_minutes, free_time_minutes, load_percentage, workload_category.

6.4 CERTIFICATIONS

Certification fact table with slowly changing dimension (SCD) support. **Key fields:** certification_id (PK), employee_id (FK), certification_date, certification_name, technology, status, expiration_date, expiration_date_indicator, validity_years, cost_usd, record_date, last_updated, latest_flag, employment_status, partner_tier, days_until_expiry, as_of_date.

7. Design Considerations

The following design principles guided architectural decisions and are embedded throughout the platform implementation:

Principle	Implementation
Auditability	GitHub Actions pipelines maintain a full commit history of all ETL runs.
Data Freshness	Automated pipeline scheduling ensures curated tables are kept up to date.
SCD Support	latest_flag and as_of_date fields in CERTIFICATIONS enable point-in-time analysis.
Governance	Semantic model enforces consistent metric definitions across all consumer queries.
Scalability	Snowflake's elastic compute scales transparently with query volume and data growth.
Usability	The Streamlit chatbot removes the need for end users to write SQL, enabling self-service analytics.